

UNIVERSITY OF MAURITIUS

FACULTY OF LAW AND MANAGEMENT



SECOND SEMESTER EXAMINATIONS

JULY / AUGUST 2020

PROGRAMME	BSc (Hons) Management (Minor: Business Informatics) Level III		
MODULE NAME	INFORMATION SYSTEMS SECURITY		
DATE	Monday 27 July 2020	MODULE CODE	CSE3216(3)
TIME	10:30 – 12:30 Hours	DURATION	2 Hours
NO. OF QUESTIONS SET	4	NO. OF QUESTIONS TO BE ATTEMPTED	4

INSTRUCTIONS TO CANDIDATES

This paper consists of FOUR (4) questions.

Answer ALL questions.

All questions carry equal marks [25 marks].

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Question 1 [25 marks]

- a) Information security means protecting information (data) and information systems from unauthorized access, use, disclosure, disruption, modification, or destruction. Information Security management is a process of defining the security controls in order to protect the information assets.
- i. Explain the three key areas used to evaluate information security in an organization. **[9 marks]**
- ii. Describe how access control can help to maintain a secure information system. Give an example. **[4 marks]**
- iii. Explain a good password policy for an organization. **[3 marks]**
- b) One of the areas of concern for a proper information security management system is the human resource component. Explain the process organizations need to put in place to minimize risk related to human resource. **[9 marks]**

Question 2 [25 marks]

- a) Briefly explain the steps involved in the first phase of conducting a security assessment for a secure information system **[8 marks]**
- b) An important concept in the design and analysis of secure systems is the security model. Using a diagram explain one security model used to protect information in a system. **[5 marks]**
- c) Differentiate between a Business Continuity Planning and a Disaster Recovery Planning. **[4 marks]**
- d) Comment on the ***Information and Communication Technologies Act*** of Mauritius. **[5 marks]**
- e) Explain green computing in terms of business ethics. **[3 marks]**

Question 3 [25 marks]

- a) Information systems are exposed to numerous threats from inside and outside the organization. One such threat is unauthorized access to the system. Explain two counter measures to protect against unauthorized access.

[6 marks]

- b) Describe two physical security controls for protecting a secure IT infrastructure.

[6 marks]

- c) Differentiate between identification, authorization and authentication method of access control in operations security.

[6 marks]

- d) To increase the level of security many organisation uses multi factor authentication. Using an appropriate example explain three factor authentication.

[7 marks]

Question 4 [25 marks]

Unprotected data, whether in transit or stored, leaves organizations vulnerable to attack, but there are effective security measures that offer robust data protection across endpoints and networks to protect data in both states.

- a) Using a diagram explain symmetric and asymmetric encryption.

[8 marks]

- b) Differentiate between:

- i. Digital certificate and security token.
- ii. Substitution cipher and transposition cipher.
- iii. Firewall and denial of service attack.

[12 marks]

- c) Many organisations use biometric system to control access to server room. Briefly explain the biometric authentication process.

[5 marks]

END OF QUESTION PAPER